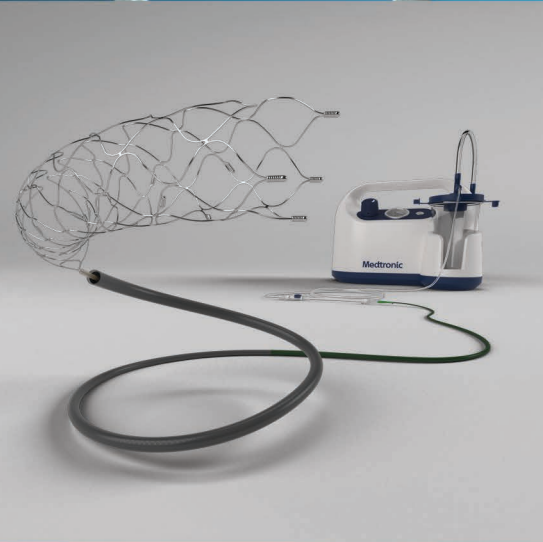




NEUROVASCULAR **2019 CATALOG**

Medtronic

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GUIDEWIRES



AVÍGO™ .014" HYDROPHILIC GUIDEWIRE

Product Catalog Number	Diameter (in)	Total Length (cm)	Coil Length (cm)
103-0606-200	.014	205	5

MIRAGE™ .008" HYDROPHILIC GUIDEWIRE

Product Catalog Number	Diameter (in)	Total Length (cm)	Coil Length (cm)
103-0608	.012>.008	200	10

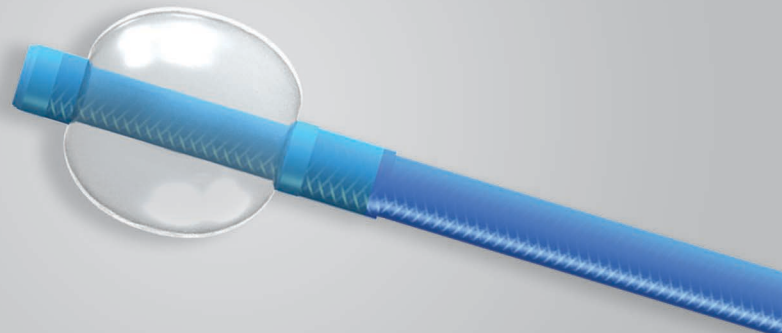
SILVERSPEED™ HYDROPHILIC GUIDEWIRE

Product Catalog Number	Diameter (in)	Total Length (cm)	Coil Length (cm)
103-0601-200	0.010	200	10
103-0602-175	0.014	175	20
103-0602-200	0.014	200	20

X-PEDION™ HYDROPHILIC GUIDEWIRE

Product Catalog Number	Diameter (in)	Total Length (cm)	Coil Length (cm)
103-0605-200	0.010	200	10
203-0602-200	0.014	200	20

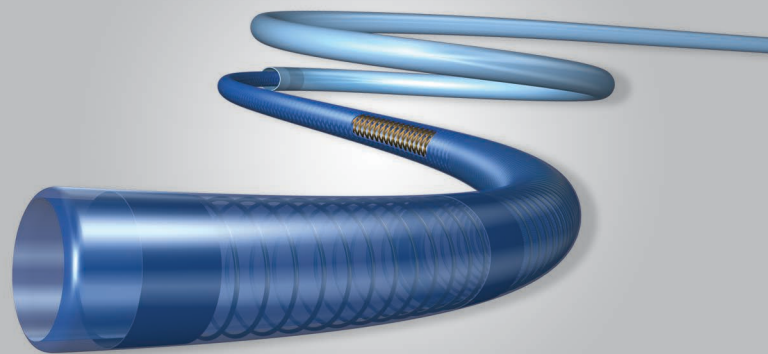
BALLOON GUIDE CATHETERS



CELLO™ BALLOON GUIDE CATHETER

Product Catalog Number	Product Name	Conformable Sheath (F)	Tip Length (mm)	Balloon Length (mm)	OD (in)	ID (cm)	Effective Length (cm)	Total Length (cm)
1610560	Cello 6F+	7	3	7	0.075	0.051	95	103
1610570	Cello 7F+	8	3	7	0.095	0.067	95	103
1610580	Cello 8F	8	3	10	0.102	0.075	95	103
1610590	Cello 9F	9	3	10	0.118	0.085	92	100

DISTAL ACCESS CATHETERS



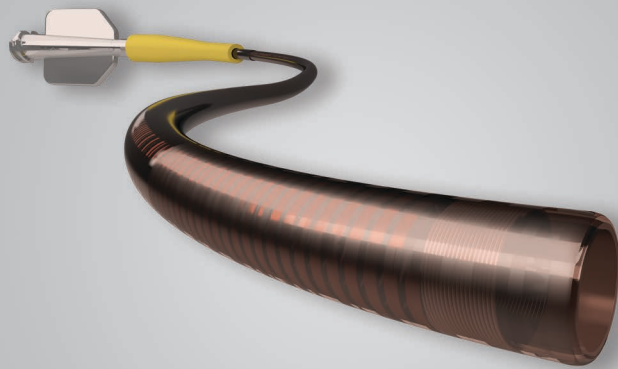
PHENOM™ PLUS CATHETER

Product Catalog Number	Working Length (cm)	Proximal Outer Diameter (in)	Distal Outer Diameter (in)	Catheter Inner Diameter (in)
FG19105-0630-1S	105	0.061	0.055	0.0445
FG19120-1030-1S	120	0.061	0.055	0.0445

REACT™ 68 CATHETER

Product Catalog Number	ID (in)	Max OD (in)	Working Length (cm)
REACT-68	0.068	0.083	132

INTRACRANIAL SUPPORT CATHETERS



NAVIEN™ INTRACRANIAL SUPPORT CATHETER

Product Catalog Number	Maximum Outer Diameter (F / in)	Inner Diameter (in)	Length (cm)	Tip Shape	Max Wire Compatibility (in)	Flexible Distal Length (cm)
RFX058-105-08	5 / 0.070	0.058	105	Straight	0.038	8
RFX058-115-08	5 / 0.070	0.058	115	Straight	0.038	8
RFX058-125-08	5 / 0.070	0.058	125	Straight	0.038	8
RFX058-130-08	5 / 0.070	0.058	130	Straight	0.038	8
RFX072-95-08	6 / 0.084	0.072	95	Straight	0.038	8
RFX072-95-08MP	6 / 0.084	0.072	95	Multi-Purpose 25°	0.038	8
RFX072-105-08	6 / 0.084	0.072	105	Straight	0.038	8
RFX072-105-08MP	6 / 0.084	0.072	105	Multi-Purpose 25°	0.038	8
RFX072-115-08	6 / 0.084	0.072	115	Straight	0.038	8
RFX072-115-08MP	6 / 0.084	0.072	115	Multi-Purpose 25°	0.038	8
RFX072-125-08	6 / 0.084	0.072	125	Straight	0.038	8
RFX072-125-08MP	6 / 0.084	0.072	125	Multi-Purpose 25°	0.039	8
RFX072-130-08	6 / 0.084	0.072	130	Straight	0.040	8
RFX072-130-08MP	6 / 0.084	0.072	130	Multi-Purpose 25°	0.041	8



MICRO CATHETERS

The **APOLLO™ ONYX™ DELIVERY MICRO CATHETER**, built on the proven Marathon™ micro catheter platform, combines exceptional navigation capabilities with a mechanically detachable tip.

APOLLO™ ONYX™ DELIVERY MICRO CATHETER

Product Catalog Number	Proximal Outer Diameter (F/in)	Distal Outer Diameter (F/in)	Inner Diameter (in)	Total Length (cm)	Tip Length (cm)	Tip Shape	Wire Compatibility (in)	Minimum Dead Space (ml)
105-5095-000	2.7 / 0.036	1.5 / 0.020	0.013	165	1.5	Straight	0.008 & 0.010 hydrophilic	0.23 ≥ 0.20 with adapter
105-5096-000	2.7 / 0.036	1.5 / 0.020	0.013	165	3.0	Straight	0.008 & 0.010 hydrophilic	0.23 ≥ 0.20 with adapter

MARATHON™ FLOW DIRECTED MICRO CATHETER

Product Catalog Number	Stylet	Outer Diameter (F)	Inner Diameter (in)	Usable Length (cm)	Distal Length (cm)	Max. Guidewire (in)
105-5056	Without	2.7>1.5	.015>.013	165	25	0.010

REBAR™ 10 REINFORCED MICRO CATHETER

Product Catalog Number	Outer Diameter (F)	Inner Diameter (in)	Usable Length (cm)	Max. Guidewire (in)
105-5078-153*	2.3>1.7	.015	153	0.012

REBAR™ 14 REINFORCED MICRO CATHETER

Product Catalog Number	Outer Diameter (F)	Inner Diameter (in)	Usable Length (cm)	Max. Guidewire (in)
105-5080-153*	2.4>1.9	.017	153	0.014

REBAR™ 18 REINFORCED MICRO CATHETER

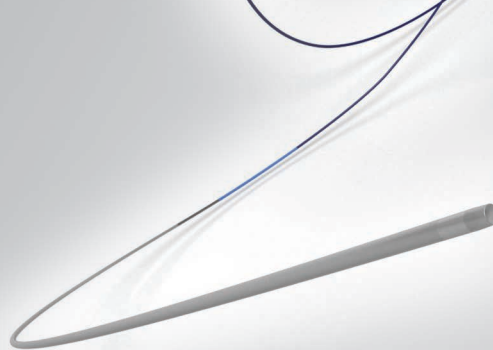
Product Catalog Number	Outer Diameter (F)	Inner Diameter (in)	Usable Length (cm)	Max. Guidewire (in)
105-5081-153*	2.7>2.4	.021	153	0.018

REBAR™ 27 REINFORCED MICRO CATHETER

Product Catalog Number	Outer Diameter (F)	Inner Diameter (in)	Usable Length (cm)	Max. Guidewire (in)
105-5082-130	2.8>2.8	.027	130	0.021

*Dual marker band

MICRO CATHETERS



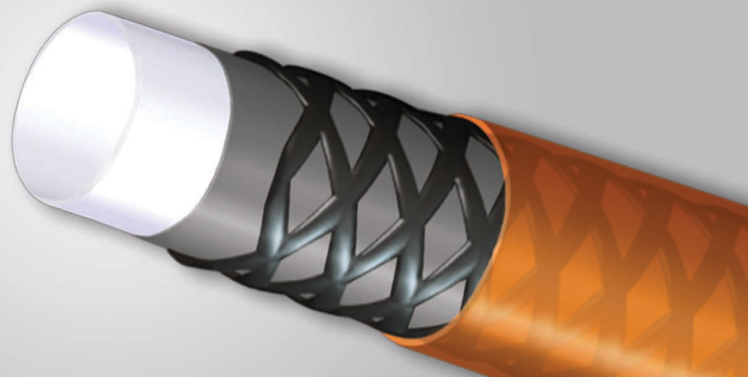
MARKSMAN™ MICRO CATHETER

Product Catalog Number	Outer Diameter (F)	Inner Diameter (in)	Usable Length (cm)	Distal Length (cm)	Max. Guidewire (in)
FA-55105-1015	3.2>2.8	0.027	105	10	0.021
FA-55135-1030	3.2>2.8	0.027	135	10	0.021
FA-55150-1030	3.2>2.8	0.027	150	10	0.021
FA-55160-1030	3.2>2.8	0.027	160	10	0.021

ORION™ -21 MICRO CATHETER

Product Catalog Number	Proximal Outer Diameter (F)	Distal Outer Diameter (F)	Inner Diameter (in)	Length (cm)	Max. Guidewire (in)	Hypotube Length (cm)
105-5098-150	2.4	2.6	0.021	150	0.018	82

MICRO CATHETERS



ECHELON™ 10 MICRO CATHETER

Product Catalog Number	Outer Diameter (F)	Inner Diameter (in)	Usable Length (cm)	Max. Guidewire (in)	Tip Configuration
105-5091-150	2.1>1.7	.017	150	0.014	Straight
145-5091-150	2.1>1.7	.017	150	0.014	45°
190-5091-150	2.1>1.7	.017	150	0.014	90°

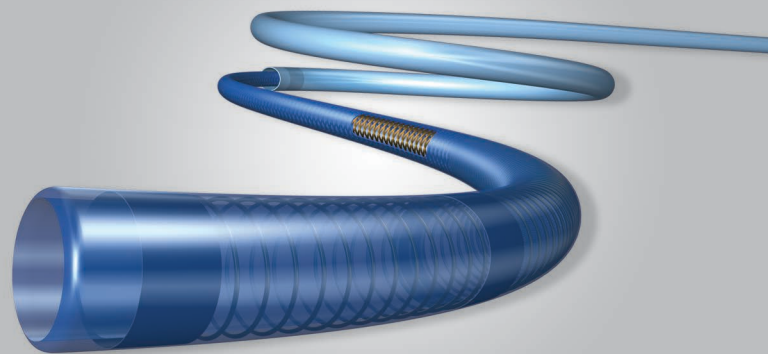
ECHELON™ 14 MICRO CATHETER

Product Catalog Number	Outer Diameter (F)	Inner Diameter (in)	Usable Length (cm)	Max. Guidewire (in)	Tip Configuration
105-5092-150	2.4>1.9	.017	150	0.014	Straight
145-5092-150	2.4>1.9	.017	150	0.014	45°
190-5092-150	2.4>1.9	.017	150	0.014	90°

ECHELON™ SYRINGE ADAPTER

Product Catalog Number	Quantity	Units/Box
103-5095	1 Box	20

MICRO CATHETERS



With features such as an elite composite shaft, proprietary inner lining, and a rounded distal tip, the **PHENOM™ FAMILY OF CATHETERS** are the ultimate delivery platform for coil and stent delivery.¹ Different clinical scenarios require variation in levels of support and navigation. With Phenom™ catheters, we provide various catheter specifications to meet your needs.

PHENOM™ 017 CATHETER

Product Catalog Number	Outer Diameter (F)	Inner Diameter (in)	Working Length (cm)	Soft Distal Segment (cm)	Flexible Single Coil Segment (cm)	Tip Shape	Max. Guidewire (in)
FG11150-0615-2S	2.2>1.8	0.017	150	6	15	Straight	0.014
FG11150-0615-2J	2.2>1.8	0.017	150	6	15	J Curve	0.014
FG11150-0615-2X	2.2>1.8	0.017	150	6	15	45 Curve	0.014
FG11150-0615-2R	2.2>1.8	0.017	150	6	15	90 Curve	0.014

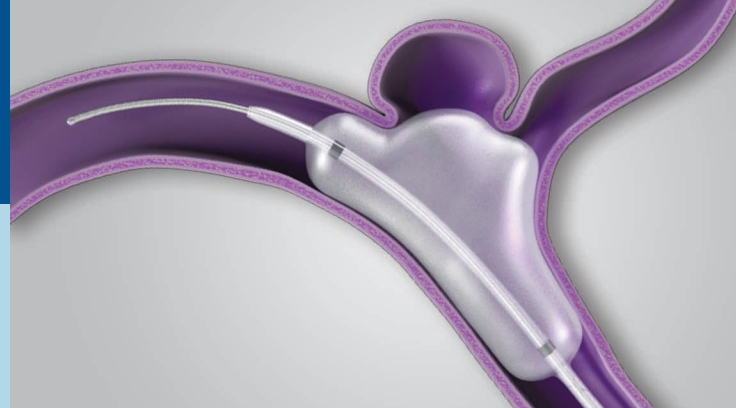
PHENOM™ 021 CATHETER

Product Catalog Number	Outer Diameter (F)	Inner Diameter (in)	Working Length (cm)	Soft Distal Segment (cm)	Flexible Single Coil Segment (cm)	Tip Shape	Max. Guidewire (in)
FG13150-0615-2S	2.6>2.3	0.021	150	6	15	Straight	0.018
FG13150-1015-2S	2.6>2.3	0.021	150	10	15	Straight	0.018

PHENOM™ 027 CATHETER

Product Catalog Number	Outer Diameter (F)	Inner Diameter (in)	Working Length (cm)	Soft Distal Segment (cm)	Flexible Single Coil Segment (cm)	Tip Shape	Max. Guidewire (in)
FG15150-0615-1S	3.1>2.8	0.027	150	6	15	Straight	0.025
FG15150-0630-1S	3.1>2.8	0.027	150	6	30	Straight	0.025

1. Cathera Document ML-0001.A
Phenom is a trademark of Cathera, Inc.



BALLOONS

HYPERFORM™ OCCLUSION BALLOON SYSTEMS

Product Catalog Number	Usable Length (cm)	Balloon Diameter (mm)	Balloon Length (mm)	Tip Length (mm)	Proximal OD (FR)	Distal OD (FR)	Guidewire (in)
104-4370	150	3	7	2	2.8	2.2	.010
104-4153	150	3	15	2	2.8	2.2	.010
104-4470	150	4	7	2	2.8	2.2	.010
104-4415	150	4	15	2	2.8	2.5	.010
104-4420	150	4	20	2	2.8	2.5	.010
104-4770	150	7	7	2	2.8	3.0	.010
104-4715	150	7	15	2	2.8	3.0	.010

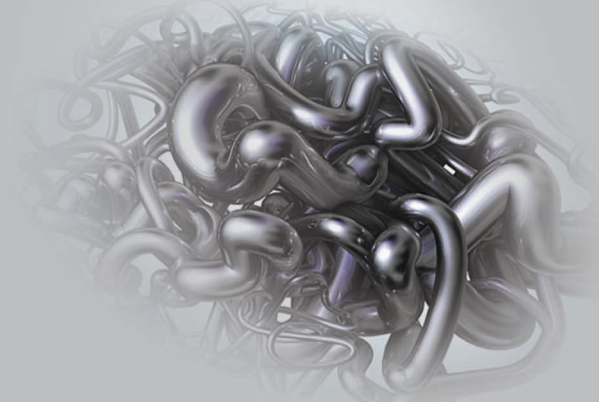
All systems packaged with an X-Pedion™ hydrophilic guidewire (103-0605-200). Balloon component not sold individually.

HYPERGLIDE™ OCCLUSION BALLOON SYSTEMS

Product Catalog Number	Usable Length (cm)	Balloon Diameter (mm)	Balloon Length (mm)	Tip Length (mm)	Proximal OD (FR)	Distal OD (FR)	Guidewire (in)
104-4310	150	3	10	4	2.8	2.2	.010
104-4315	150	3	15	4	2.8	2.2	.010
104-4113	150	4	10	4	2.8	2.2	.010
104-4112	150	4	15	4	2.8	2.2	.010
104-4127	150	4	20	4	2.8	2.2	.010
104-4132	150	4	30	4	2.8	2.2	.010
104-4515	150	5	15	4	2.8	2.2	.010
104-4520	150	5	20	4	2.8	2.2	.010
104-4530	150	5	30	4	2.8	2.2	.010

All systems packaged with an X-Pedion™ hydrophilic guidewire (103-0605-200). Balloon component not sold individually.

LIQUID EMBOLICS



ONYX™ LIQUID EMBOLIC SYSTEM provides the advantage of time and the power of control for presurgical embolization of brain arterio-venous malformations (AVMs).¹

ONYX™ LIQUID EMBOLIC SYSTEM

Product Catalog Number	Description
105-7100-060	Onyx™ 18 LES
105-7100-080	Onyx™ 34 LES

The Onyx™ 18 and 34 LES Kits contain:

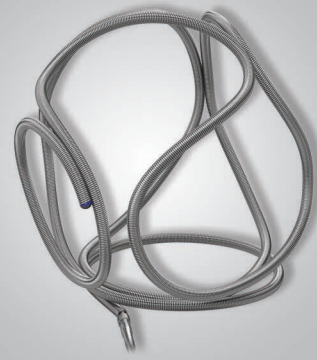
- 1.5 ml vial of Onyx™ LES (1)
- 1.5 ml vial of DMSO (1)
- 1 ml DMSO syringe (1)
- 1 ml Onyx™ LES syringe (2)

ONYX™ LES ACCESSORIES

Product Catalog Number	Description
103-1205-001	Vial Mixer

1. IFU 70753 - OCI Rev. 03/14

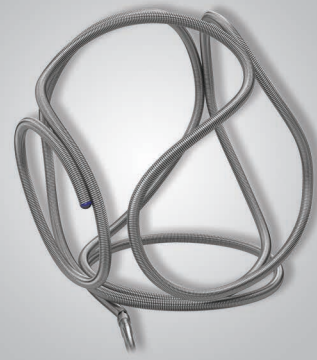
3D FRAMING DETACHABLE COILS



AXIUM™ PRIME DETACHABLE COILS (FRAME)

Product Catalog Number	Diameter (mm)	Length (cm)	Progressive Coil Diameter (in)
FC-3-6-3D	3	6	0.0115
FC-3-8-3D	3	8	0.0115
FC-3-10-3D	3	10	0.0115
FC-3.5-6-3D	3.5	6	0.0115
FC-3.5-8-3D	3.5	8	0.0115
FC-3.5-10-3D	3.5	10	0.0115
FC-4-6-3D	4	6	0.0125
FC-4-8-3D	4	8	0.0125
FC-4-10-3D	4	10	0.0125
FC-4-12-3D	4	12	0.0125
FC-4-15-3D	4	15	0.0125
FC-5-8-3D	5	8	0.0125
FC-5-10-3D	5	10	0.0125
FC-5-15-3D	5	15	0.0125
FC-5-20-3D	5	20	0.0125
FC-6-10-3D	6	10	0.0125
FC-6-15-3D	6	15	0.0125
FC-6-20-3D	6	20	0.0125
FC-6-25-3D	6	25	0.0125
FC-7-12-3D	7	12	0.0135
FC-7-15-3D	7	15	0.0135
FC-7-20-3D	7	20	0.0135
FC-7-30-3D	7	30	0.0135
FC-8-15-3D	8	15	0.0135
FC-8-20-3D	8	20	0.0135
FC-8-30-3D	8	30	0.0135
FC-9-20-3D	9	20	0.0135
FC-9-30-3D	9	30	0.0135

3D FRAMING DETACHABLE COILS



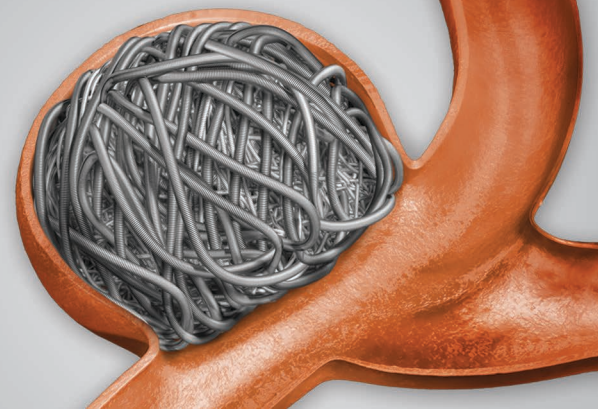
AXIUM™ PRIME DETACHABLE COILS (FRAME)

Product Catalog Number	Diameter (mm)	Length (cm)	Progressive Coil Diameter (in)
FC-10-20-3D	10	20	0.0135
FC-10-30-3D	10	30	0.0135
FC-10-40-3D	10	40	0.0135
FC-12-30-3D	12	30	0.0145
FC-12-40-3D	12	40	0.0145
FC-12-50-3D	12	50	0.0145
FC-14-30-3D	14	30	0.0145
FC-14-40-3D	14	40	0.0145
FC-14-50-3D	14	50	0.0145
FC-16-40-3D	16	40	0.0145
FC-16-50-3D	16	50	0.0145
FC-18-40-3D	18	40	0.0145
FC-18-50-3D	18	50	0.0145
FC-20-50-3D	20	50	0.0145
FC-22-50-3D	22	50	0.0145
FC-25-50-3D	25	50	0.0145

ID INSTANT DETACHER (5 PACK)

Product Catalog Number	Description
ID-1-5	Axium™ ID Instant Detacher

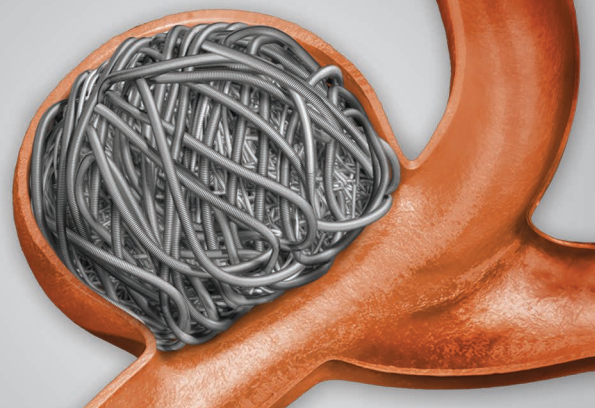
3D EXTRA SOFT DETACHABLE COILS



AXIUM™ PRIME 3D DETACHABLE COILS (EXTRA SOFT)

Product Catalog Number	Diameter (mm)	Length (cm)	Progressive Coil Diameter (in)
APB-1-2-3D-ES	1	2	0.0108
APB-1-3-3D-ES	1	3	0.0108
APB-1-4-3D-ES	1	4	0.0108
APB-1.5-2-3D-ES	1.5	2	0.0108
APB-1.5-3-3D-ES	1.5	3	0.0108
APB-1.5-4-3D-ES	1.5	4	0.0108
APB-2-2-3D-ES	2	2	0.0108
APB-2-3-3D-ES	2	3	0.0108
APB-2-4-3D-ES	2	4	0.0108
APB-2.5-4-3D-ES	2.5	4	0.0108
APB-2.5-6-3D-ES	2.5	6	0.0108
APB-3-4-3D-ES	3	4	0.0108
APB-3-6-3D-ES	3	6	0.0108
APB-3-8-3D-ES	3	8	0.0108
APB-3.5-6-3D-ES	3.5	6	0.0108
APB-3.5-8-3D-ES	3.5	8	0.0108
APB-3.5-10-3D-ES	3.5	10	0.0108

HELICAL EXTRA SOFT DETACHABLE COILS



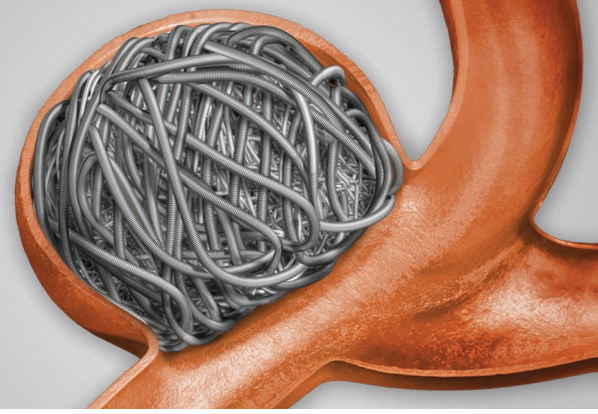
AXIUM™ PRIME HELICAL DETACHABLE COILS (EXTRA SOFT)

Product Catalog Number	Diameter (mm)	Length (cm)	Progressive Coil Diameter (in)
APB-1-1-HX-ES	1	1	0.0108
APB-1-2-HX-ES	1	2	0.0108
APB-1-3-HX-ES	1	3	0.0108
APB-1.5-1-HX-ES	1.5	1	0.0108
APB-1.5-2-HX-ES	1.5	2	0.0108
APB-1.5-3-HX-ES	1.5	3	0.0108
APB-1.5-4-HX-ES	1.5	4	0.0108
APB-2-1-HX-ES	2	1	0.0108
APB-2-2-HX-ES	2	2	0.0108
APB-2-3-HX-ES	2	3	0.0108
APB-2-4-HX-ES	2	4	0.0108
APB-2-6-HX-ES	2	6	0.0108
APB-2-8-HX-ES	2	8	0.0108
APB-2.5-3-HX-ES	2.5	3	0.0108
APB-2.5-4-HX-ES	2.5	4	0.0108
APB-2.5-6-HX-ES	2.5	6	0.0108
APB-2.5-8-HX-ES	2.5	8	0.0108
APB-3-4-HX-ES	3	4	0.0108
APB-3-6-HX-ES	3	6	0.0108
APB-3-8-HX-ES	3	8	0.0108
APB-3-10-HX-ES	3	10	0.0108

ID INSTANT DETACHER (5 PACK)

Product Catalog Number	Description
ID-1-5	Axium™ ID Instant Detacher

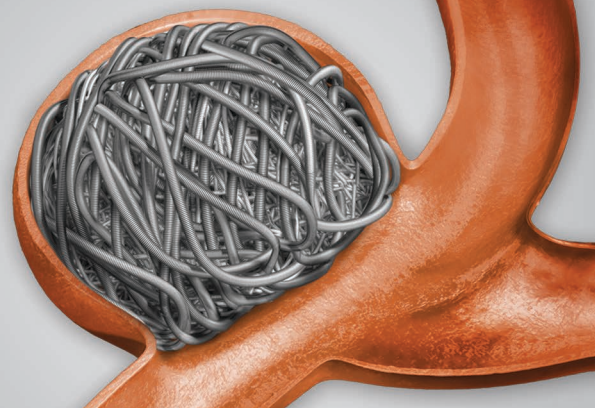
3D SUPER SOFT DETACHABLE COILS



AXIUM™ PRIME 3D DETACHABLE COILS (SUPER SOFT)

Product Catalog Number	Diameter (mm)	Length (cm)	Progressive Coil Diameter (in)
APB-4-6-3D-SS	4	6	0.0115
APB-4-8-3D-SS	4	8	0.0115
APB-4-10-3D-SS	4	10	0.0115
APB-4-12-3D-SS	4	12	0.0115
APB-5-8-3D-SS	5	8	0.0115
APB-5-10-3D-SS	5	10	0.0115
APB-5-15-3D-SS	5	15	0.0115
APB-6-10-3D-SS	6	10	0.0115
APB-6-15-3D-SS	6	15	0.0115
APB-6-20-3D-SS	6	20	0.0115

HELICAL SUPER SOFT DETACHABLE COILS



AXIUM™ PRIME HELICAL DETACHABLE COILS (SUPER SOFT)

Product Catalog Number	Diameter (mm)	Length (cm)	Progressive Coil Diameter (in)
APB-4-6-HX-SS	4	6	0.0115
APB-4-8-HX-SS	4	8	0.0115
APB-4-10-HX-SS	4	10	0.0115
APB-4-12-HX-SS	4	12	0.0115
APB-5-10-HX-SS	5	10	0.0115
APB-5-15-HX-SS	5	15	0.0115
APB-5-20-HX-SS	5	20	0.0115
APB-6-12-HX-SS	6	12	0.0115
APB-6-20-HX-SS	6	20	0.0115

ID INSTANT DETACHER (5 PACK)

Product Catalog Number	Description
ID-1-5	Axium™ ID Instant Detacher

3D DETACHABLE COILS



AXIUM™ 3D DETACHABLE COILS

Product Catalog Number	Diameter (mm)	Length (cm)	Progressive Coil Diameter (in)
QC-2-2-3D	2	2	0.0115
QC-2-4-3D	2	4	0.0115
QC-2-6-3D	2	6	0.0115
QC-2.5-2-3D	2.5	2	0.0115
QC-2.5-4-3D	2.5	4	0.0115
QC-2.5-6-3D	2.5	6	0.0115
QC-2.5-8-3D	2.5	8	0.0115
QC-3-4-3D	3	4	0.0115
QC-3-6-3D	3	6	0.0115
QC-3-8-3D	3	8	0.0115
QC-3-10-3D	3	10	0.0115
QC-3.5-6-3D	3.5	6	0.0115
QC-3.5-12-3D	3.5	12	0.0115
QC-3.5-15-3D	3.5	15	0.0115
QC-4-6-3D	4	6	0.0125
QC-4-8-3D	4	8	0.0125
QC-4-10-3D	4	10	0.0125
QC-4-12-3D	4	12	0.0125
QC-5-8-3D	5	8	0.0125
QC-5-10-3D	5	10	0.0125
QC-5-15-3D	5	15	0.0125
QC-6-10-3D	6	10	0.0125
QC-6-15-3D	6	15	0.0125
QC-6-20-3D	6	20	0.0125

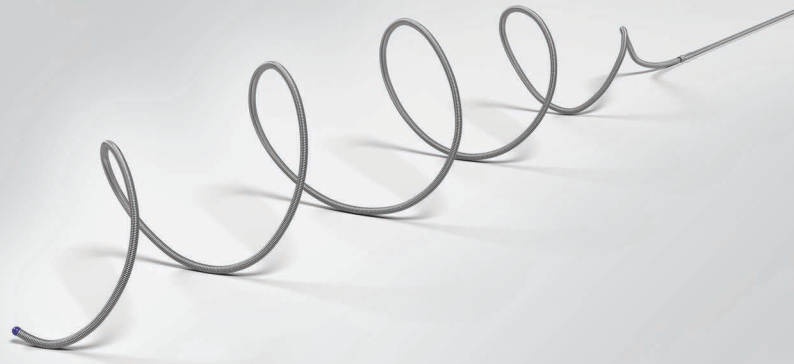
3D DETACHABLE COILS



AXIUM™ 3D DETACHABLE COILS

Product Catalog Number	Diameter (mm)	Length (cm)	Progressive Coil Diameter (in)
QC-7-15-3D	7	15	0.0135
QC-7-20-3D	7	20	0.0135
QC-7-30-3D	7	30	0.0135
QC-8-15-3D	8	15	0.0135
QC-8-20-3D	8	20	0.0135
QC-8-30-3D	8	30	0.0135
QC-9-20-3D	9	20	0.0135
QC-9-30-3D	9	30	0.0135
QC-10-20-3D	10	20	0.0135
QC-10-30-3D	10	30	0.0135
QC-12-30-3D	12	30	0.0145
QC-12-40-3D	12	40	0.0145
QC-14-30-3D	14	30	0.0145
QC-14-40-3D	14	40	0.0145
QC-16-40-3D	16	40	0.0145
QC-18-40-3D	18	40	0.0145
QC-20-50-3D	20	50	0.0145
QC-22-50-3D	22	50	0.0145
QC-25-50-3D	25	50	0.0145

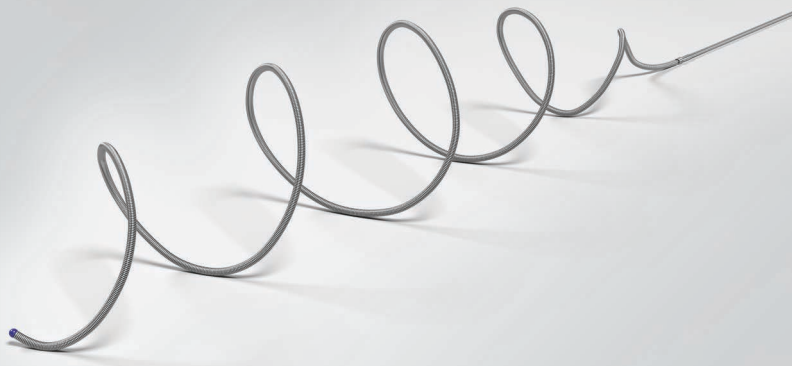
HELICAL DETACHABLE COILS



AXIUM™ HELICAL DETACHABLE COILS

Product Catalog Number	Diameter (mm)	Length (cm)	Progressive Coil Diameter (in)
QC-1.5-1-HELIX	1.5	1	0.0115
QC-1.5-2-HELIX	1.5	2	0.0115
QC-1.5-3-HELIX	1.5	3	0.0115
QC-1.5-4-HELIX	1.5	4	0.0115
QC-2-1-HELIX	2	1	0.0115
QC-2-2-HELIX	2	2	0.0115
QC-2-3-HELIX	2	3	0.0115
QC-2-4-HELIX	2	4	0.0115
QC-2-6-HELIX	2	6	0.0115
QC-2-8-HELIX	2	8	0.0115
QC-2.5-2-HELIX	2.5	2	0.0115
QC-2.5-4-HELIX	2.5	4	0.0115
QC-2.5-6-HELIX	2.5	6	0.0115
QC-2.5-8-HELIX	2.5	8	0.0115
QC-3-4-HELIX	3	4	0.0115
QC-3-6-HELIX	3	6	0.0115
QC-3-8-HELIX	3	8	0.0115
QC-4-8-HELIX	4	8	0.0125
QC-4-10-HELIX	4	10	0.0125
QC-4-12-HELIX	4	12	0.0125
QC-5-15-HELIX	5	15	0.0125
QC-5-20-HELIX	5	20	0.0125
QC-6-20-HELIX	6	20	0.0125

HELICAL DETACHABLE COILS



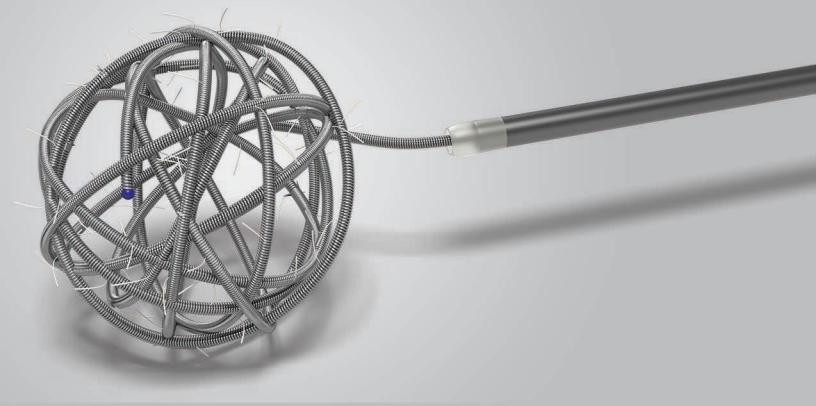
AXIUM™ HELICAL DETACHABLE COILS

Product Catalog Number	Diameter (mm)	Length (cm)	Progressive Coil Diameter (in)
QC-7-20-HELIX	7	20	0.0135
QC-7-30-HELIX	7	30	0.0135
QC-8-20-HELIX	8	20	0.0135
QC-8-30-HELIX	8	30	0.0135
QC-9-20-HELIX	9	20	0.0135
QC-9-30-HELIX	9	30	0.0135
QC-10-20-HELIX	10	20	0.0135
QC-10-30-HELIX	10	30	0.0135
QC-12-30-HELIX	12	30	0.0145
QC-12-40-HELIX	12	40	0.0145
QC-14-30-HELIX	14	30	0.0145
QC-14-40-HELIX	14	40	0.0145
QC-16-30-HELIX	16	30	0.0145
QC-16-40-HELIX	16	40	0.0145
QC-18-40-HELIX	18	40	0.0145
QC-20-40-HELIX	20	40	0.0145
QC-20-50-HELIX	20	50	0.0145

ID INSTANT DETACHER (5 PACK)

Product Catalog Number	Description
ID-1-5	Axium™ ID Instant Detacher

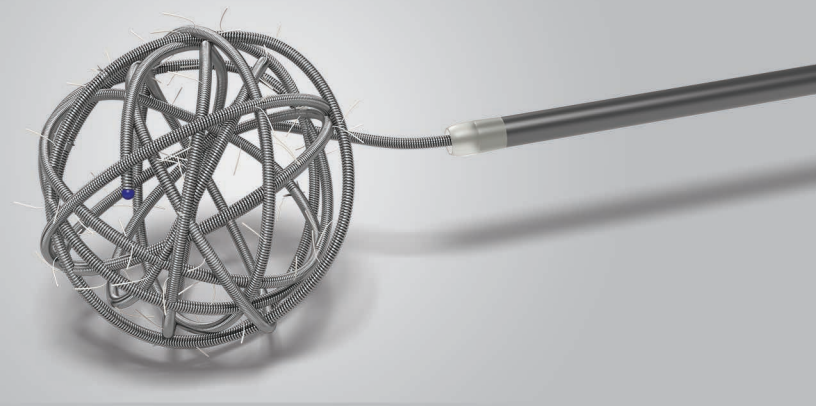
PGLA 3D COILS



AXIUM™ MICROFX™ PGLA 3D COILS

Product Catalog Number	Diameter (mm)	Length (cm)	Progressive Coil Diameter (in)
PC-2-2-3D	2	2	0.0115
PC-2-4-3D	2	4	0.0115
PC-2-6-3D	2	6	0.0115
PC-3-4-3D	3	4	0.0115
PC-3-6-3D	3	6	0.0115
PC-3-8-3D	3	8	0.0115
PC-4-6-3D	4	6	0.0125
PC-4-8-3D	4	8	0.0125
PC-4-10-3D	4	10	0.0125
PC-4-12-3D	4	12	0.0125
PC-5-8-3D	5	8	0.0125
PC-5-10-3D	5	10	0.0125
PC-5-15-3D	5	15	0.0125
PC-6-10-3D	6	10	0.0125
PC-6-15-3D	6	15	0.0125
PC-6-20-3D	6	20	0.0125
PC-7-15-3D	7	15	0.0135
PC-7-20-3D	7	20	0.0135
PC-7-30-3D	7	30	0.0135
PC-8-15-3D	8	15	0.0135
PC-8-20-3D	8	20	0.0135

PGLA 3D COILS



AXIUM™ MICROFX™ PGLA 3D COILS

Product Catalog Number	Diameter (mm)	Length (cm)	Progressive Coil Diameter (in)
PC-8-30-3D	8	30	0.0135
PC-9-20-3D	9	20	0.0135
PC-9-30-3D	9	30	0.0135
PC-10-20-3D	10	20	0.0135
PC-10-30-3D	10	30	0.0135
PC-12-30-3D	12	30	0.0145
PC-12-40-3D	12	40	0.0145
PC-14-30-3D	14	30	0.0145
PC-14-40-3D	14	40	0.0145
PC-16-40-3D	16	40	0.0145
PC-18-40-3D	18	40	0.0145

PGLA HELICAL COILS



AXIUM™ MICROFX™ PGLA HELIX COILS

Product Catalog Number	Diameter (mm)	Length (cm)	Progressive Coil Diameter (in)
PC-2-1-HELIX	2	1	0.0115
PC-2-2-HELIX	2	2	0.0115
PC-2-3-HELIX	2	3	0.0115
PC-2-4-HELIX	2	4	0.0115
PC-2-6-HELIX	2	6	0.0115
PC-2-8-HELIX	2	8	0.0115
PC-3-4-HELIX	3	4	0.0115
PC-3-6-HELIX	3	6	0.0115
PC-3-8-HELIX	3	8	0.0115
PC-4-8-HELIX	4	8	0.0125
PC-4-10-HELIX	4	10	0.0125
PC-4-12-HELIX	4	12	0.0125
PC-5-15-HELIX	5	15	0.0125
PC-5-20-HELIX	5	20	0.0125
PC-6-20-HELIX	6	20	0.0125
PC-7-20-HELIX	7	20	0.0135
PC-7-30-HELIX	7	30	0.0135
PC-8-20-HELIX	8	20	0.0135
PC-8-30-HELIX	8	30	0.0135
PC-9-20-HELIX	9	20	0.0135
PC-9-30-HELIX	9	30	0.0135
PC-10-20-HELIX	10	20	0.0135
PC-10-30-HELIX	10	30	0.0135

NYLON HELICAL COILS



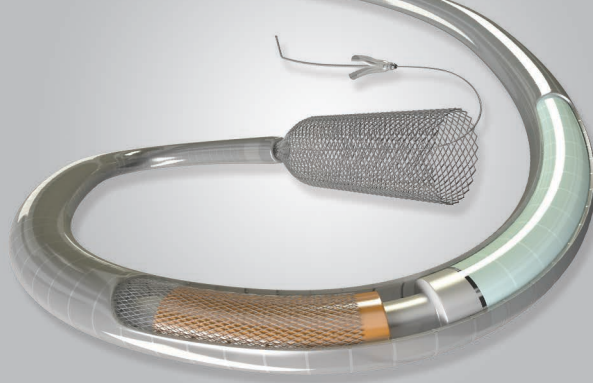
AXIUM™ MICROFX™ NYLON COILS

Product Catalog Number	Diameter (mm)	Length (cm)	Progressive Coil Diameter (in)
NC-2-1-HELIX	2	1	0.0115
NC-2-2-HELIX	2	2	0.0115
NC-2-3-HELIX	2	3	0.0115
NC-2-4-HELIX	2	4	0.0115
NC-2-6-HELIX	2	6	0.0115
NC-2-8-HELIX	2	8	0.0115
NC-3-4-HELIX	3	4	0.0115
NC-3-6-HELIX	3	6	0.0115
NC-3-8-HELIX	3	8	0.0115
NC-4-8-HELIX	4	8	0.0125
NC-4-10-HELIX	4	10	0.0125

ID INSTANT DETACHER (5 PACK)

Product Catalog Number	Description
ID-1-5	Axium™ ID Instant Detacher

FLOW DIVERTERS



The **PIPELINE™ FLEX EMBOLIZATION DEVICE** features a clinically proven braid design for flexible conformity, and a redesigned, softer distal tip.¹ It's trackable through tortuosity, enables better vessel deflection, and allows for a controlled distal landing zone.² The Pipeline™ device is fully resheathable and can be repositioned and redeployed up to two times.³

PIPELINE™ FLEX EMBOLIZATION DEVICE

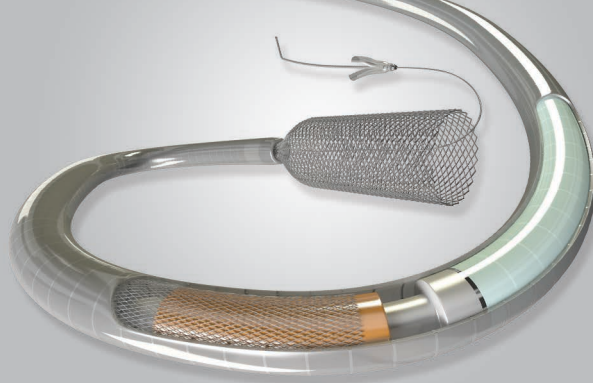
Product Catalog Number	Diameter (mm)	Length (mm)
PED-250-10	2.50	10
PED-250-12	2.50	12
PED-250-14	2.50	14
PED-250-16	2.50	16
PED-250-18	2.50	18
PED-250-20	2.50	20
PED-275-10	2.75	10
PED-275-12	2.75	12
PED-275-14	2.75	14
PED-275-16	2.75	16
PED-275-18	2.75	18
PED-275-20	2.75	20
PED-300-10	3.00	10
PED-300-12	3.00	12
PED-300-14	3.00	14
PED-300-16	3.00	16
PED-300-18	3.00	18
PED-300-20	3.00	20
PED-300-25	3.00	25
PED-300-30	3.00	30
PED-300-35	3.00	35
PED-325-10	3.25	10
PED-325-12	3.25	12
PED-325-14	3.25	14
PED-325-16	3.25	16
PED-325-18	3.25	18
PED-325-20	3.25	20
PED-325-25	3.25	25
PED-325-30	3.25	30

1. TR-NV10931/TR-NV11121. Rev. A.

2. TR-NV10931 Rev. A/TR-NV11121 Rev. A

3. TR-NV11534/TR-NV11121. Rev. A.

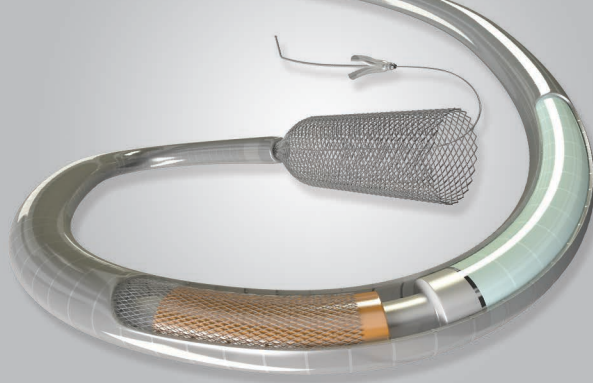
FLOW DIVERTERS



PIPELINE™ FLEX EMBOLIZATION DEVICE

Product Catalog Number	Diameter (mm)	Length (mm)
PED-325-35	3.25	35
PED-350-10	3.50	10
PED-350-12	3.50	12
PED-350-14	3.50	14
PED-350-16	3.50	16
PED-350-18	3.50	18
PED-350-20	3.50	20
PED-350-25	3.50	25
PED-350-30	3.50	30
PED-350-35	3.75	35
PED-375-10	3.75	10
PED-375-12	3.75	12
PED-375-14	3.75	14
PED-375-16	3.75	16
PED-375-18	3.75	18
PED-375-20	3.75	20
PED-375-25	3.75	25
PED-375-30	3.75	30
PED-375-35	4.00	35
PED-400-10	4.00	10
PED-400-12	4.00	12
PED-400-14	4.00	14
PED-400-16	4.00	16
PED-400-18	4.00	18
PED-400-20	4.00	20
PED-400-25	4.00	25
PED-400-30	4.00	30
PED-400-35	4.00	35
PED-425-10	4.25	10
PED-425-12	4.25	12
PED-425-14	4.25	14
PED-425-16	4.25	16

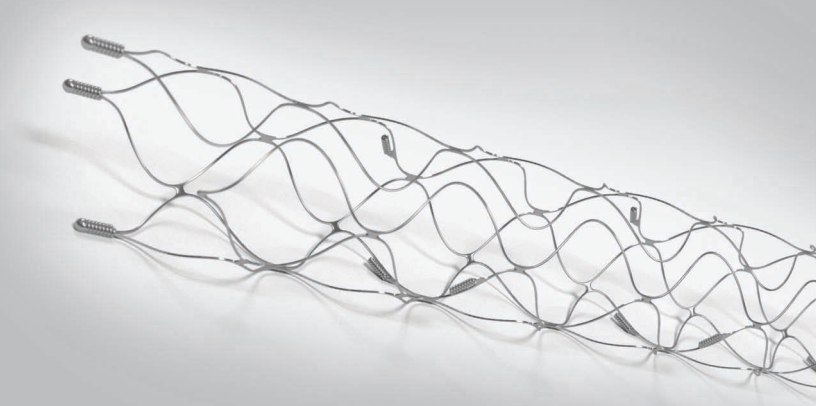
FLOW DIVERTERS



PIPELINE™ FLEX EMBOLIZATION DEVICE

Product Catalog Number	Diameter (mm)	Length (mm)
PED-425-18	4.25	18
PED-425-20	4.25	20
PED-425-25	4.25	25
PED-425-30	4.25	30
PED-425-35	4.25	35
PED-450-10	4.50	10
PED-450-12	4.50	12
PED-450-14	4.50	14
PED-450-16	4.50	16
PED-450-18	4.50	18
PED-450-20	4.50	20
PED-450-25	4.50	25
PED-450-30	4.50	30
PED-450-35	4.50	35
PED-475-10	4.75	10
PED-475-12	4.75	12
PED-475-14	4.75	14
PED-475-16	4.75	16
PED-475-18	4.75	18
PED-475-20	4.75	20
PED-475-25	4.75	25
PED-475-30	4.75	30
PED-475-35	4.75	35
PED-500-10	5.00	10
PED-500-12	5.00	12
PED-500-14	5.00	14
PED-500-16	5.00	16
PED-500-18	5.00	18
PED-500-20	5.00	20
PED-500-25	5.00	35
PED-500-30	5.00	30
PED-500-35	5.00	35

REVASCULARIZATION DEVICES



The **SOLITAIRE™ PLATINUM REVASCULARIZATION DEVICE** is a mechanical thrombectomy device designed to restore blood flow by removing thrombus to reduce disability in patients experiencing ischemic stroke due to large intracranial vessel occlusion. This device, with the addition of platinum markers, is designed to provide real time procedural feedback¹ when being used in the neurovasculature such as the Internal Carotid Artery, M1 and M2 segments of the Middle Cerebral Artery, Anterior Cerebral Artery, Basilar Artery, and Vertebral Arteries.²

SOLITAIRE™ PLATINUM REVASCULARIZATION DEVICE

Product Catalog Number	Size (mm)	Recommended Vessel Diameter (mm)	Minimum Micro Catheter ID (in)	Push Wire Length (cm)	# of Radiopaque Markers			Radiopaque Marker Spacing (mm)	Total # of Radiopaque Marker Rows
					Distal end	Proximal end	Stent body		
SFR3-4-20-05	4 x 20	2.0 – 4.0	0.021	180	3	1	12	5	5
SFR3-4-20-10	4 x 20	2.0 – 4.0	0.021	180	3	1	3	10	3
SFR3-4-40-10	4 x 40	2.0 – 4.0	0.021	180	3	1	3	10	5
SFR3-6-20-10	6 x 20	3.0 – 5.5	0.027	180	4	1	3	10	3
SFR3-6-24-06	6 x 24	3.0 – 5.5	0.027	180	4	1	12	6	5
SFR3-6-40-10	6 x 40	3.0 – 5.5	0.027	180	4	1	12	10	5

1. TR-NV12692 rev A.
2. 71011-001 Rev. 06/17.

ASPIRATION



RIPTIDE™ ASPIRATION SYSTEM

					Product Catalog Number								
Riptide™ Aspiration Pump					MAP-1000								
				Product Catalog Number		Volume							
Riptide™ Collection Canister & Intermediate Tubing				MAC-1200		1200mL							
		Product Catalog Number		Inner Diameter		Tubing Length		Distal Length					
Riptide™ Aspiration Tubing		AT-88-110		0.088"		112"		7"					
		Product Catalog Number		Working Length		Prox Max OD		Dist Max OD		Prox ID		Dist ID	
React™ 68 Catheter		REACT-68		132cm		0.083"		0.083"		0.068"		0.068"	

ACCESSORIES

The **CADENCE™ PRECISION INJECTOR** provides precision inflation of balloon catheters; 0.02 ml inflation per rotation offers tactile feedback.

CADENCE™ PRECISION INJECTOR ACCESSORY		
Product Catalog Number	Capacity (ml)	Quantity
103-0304	1	5

1 ML INJECTION SYRINGE		
Product Catalog Number	Capacity (ml)	Syringes/Box
103-1203	1	10

INDICATIONS FOR USE

Specifications Nominal

Indications, contraindications, warnings and instructions for use can be found in the product labeling supplied with each device. **CAUTION:** Federal (U.S.A.) law restricts this device to sale by or on the order of a physician.

Avigo™ hydrophilic guidewire

The Avigo™ hydrophilic guidewire is indicated for general intravascular use to aid in the selective placement of catheters in the peripheral and cerebral vasculature during diagnostic and/or therapeutic procedures. The device is not intended for use in the coronary arteries.

Mirage™ and SilverSpeed™ hydrophilic guidewire

The Mirage™ and SilverSpeed™ hydrophilic guidewires are indicated for general intravascular use to aid in the selective placement of catheters in the peripheral, visceral and cerebral vasculature during diagnostic and/or therapeutic procedures.

X-Pedion™ hydrophilic guidewire

The X-Pedion™ hydrophilic guidewire is indicated for general intravascular use to aid in the selective placement of catheters in the peripheral, visceral and cerebral vasculature during diagnostic and/or therapeutic procedures.

Phenom™ Plus and Phenom™ catheter

Phenom™ Catheters are intended for the introduction of interventional devices and infusion of diagnostic or therapeutic agents into the neuro, peripheral, and coronary vasculatures.

React™ 68 Catheter

The React™ 68 Catheter is indicated for the introduction of interventional devices into the peripheral and neuro vasculature.

Cello™ balloon guide catheter

The Cello™ balloon guide catheter is indicated for use in facilitating the insertion and guidance of intravascular catheters into a selected blood vessel in the peripheral and neuro vascular systems. The balloon provides temporary vascular occlusion during these and other angiographic procedures.

Cello™ is a trademark of and is manufactured by Fuji Systems Corporation.

Navien™ intracranial support catheter

The Navien™ intracranial support catheter is indicated for the introduction of interventional devices into the peripheral and neuro vasculature.

Apollo™ Onyx™ delivery micro catheter

This product is for the exclusive use by medical specialists experienced in angiographic and percutaneous neurointerventional procedures.

Indications For Use: The Apollo™ Onyx™ delivery micro catheter is intended to access the neuro vasculature for the controlled selective infusion of the Onyx™ liquid embolic system (LES).

Contraindications:

- The Apollo™ Onyx™ delivery micro catheter is contraindicated when, in the medical judgment of the physician, use of such product may compromise the patient's condition.
- Not intended for use in the coronary vasculature.

Precautions: 1) Select tip size based on angioarchitecture. The detachment zone should never be distal to the last tortuous curve of the vessel. Refluxing over the detachment zone distal to the last tortuous curve may result in catheter entrapment. Do not place catheter such that the detached tip could interfere with patent vessels. 2) Prior to use, carefully examine the Apollo™ Onyx™ delivery micro catheter and its packaging to verify that it has not been damaged during shipment. Do not touch or manipulate the catheter tip prior to use. 3) Prior to use, all accessory devices and agents should be fully prepared according to the manufacturer's instructions. 4) During navigation, check that the distal tip of the catheter is not kinked before passing the guidewire through it. Kinking or prolapsing of the catheter may result in unintended rupture of the catheter. 5) Always monitor infusion rates when using the catheter. 6) The Apollo™ Onyx™ delivery micro catheter has a hydrophilic coating on the outside of the catheter which must be kept hydrated. 7) This catheter is not intended for use with chemotherapy agents. 8) When the infusion catheter is in the body, it should be manipulated only under fluoroscopy. Do not attempt to move the catheter without observing the resultant tip response. 9) Navigating or repositioning the catheter while it is in a wedged position or with vessels that are in vasospasm may cause premature tip detachment. 10) When performing angiography, it is recommended to use a 3cc syringe rather than a 1cc syringe to reduce the risk of catheter over-pressurization. 11) The Apollo™ Onyx™ delivery micro catheter is a flow directed micro catheter that can optionally be used with hydrophilic, 0.010" or less sized guidewires. The Apollo™ Onyx™ delivery micro catheter is not compatible with non-hydrophilically coated guidewires or guidewires greater than 0.010" in diameter. 12) It is recommended that the Apollo™ Onyx™ delivery micro catheter be used with an appropriately sized guiding catheter which allows adequate clearance (minimum internal diameter of 0.053" or 1.35mm). 13) When withdrawing the catheter, monitor the distal tip under angiography. Pulling the catheter against significant resistance can cause patient injury. If significant catheter resistance is felt, refer to the precaution in the procedure section of the Instructions for Use provided with the device for guidance. 14) If catheter entrapment is suspected, fast catheter retrieval technique may result in catheter shaft separation and potential vascular damage. Follow catheter retrieval instructions at the end of instructions for use.

Potential Complications: Potential complications include, but are not limited to: Puncture site hematoma, Vessel perforation, Vessel spasm, Hemorrhage, Pain and tenderness, Thrombolytic episodes, Neurological deficits including stroke and death, Vascular thrombosis.

Marathon™ micro catheter

The Marathon™ micro catheter is intended to access peripheral and neuro vasculature for the controlled selective infusion of physician-specified therapeutic agents such as embolization materials and of diagnostic materials such as contrast media. Not intended for use in the coronary vasculature.

Marksman™ micro catheter

The Marksman™ micro catheter is intended for the introduction of interventional devices and infusion of diagnostic or therapeutic agents into the neuro, peripheral, and coronary vasculature.

INDICATIONS FOR USE

Orion™ -21 micro catheter

The Orion™ -21 micro catheter is intended for the controlled selective infusion of physician-specified therapeutic agents or contrast media into the vasculature of the peripheral and neuro anatomy.

Echelon™ micro catheter

The Echelon™ micro catheter is intended to access peripheral and neuro vasculature for the controlled selective infusion of physician-specified therapeutic agents such as embolization materials and of diagnostic materials such as contrast media.

Rebar™ micro catheter

The Rebar™ micro catheter is intended for the controlled selective infusion of physician-specified therapeutic agents or contrast media into the vasculature of the peripheral and neuro anatomy.

Hyperform™ and HyperGlide™ occlusion balloon system

The Hyperform™ and HyperGlide™ occlusion balloon catheters are indicated for use in blood vessels of the peripheral and neuro vasculature where temporary occlusion is desired. These catheters offer a vessel selective technique of temporary vascular occlusion, which is useful in selectively stopping or controlling blood flow; the occlusion balloon catheters may also be used in balloon-assisted embolization of intracranial aneurysms.

Onyx™ liquid embolic system (AVM)

This product is for the exclusive use by medical specialists experienced in angiographic and percutaneous neurointerventional procedures.

Indications For Use: Presurgical embolization of brain arteriovenous malformations (bAVMs).

Contraindications: The use of the Onyx™ LES is contraindicated when any of the following conditions exist:

- When optimal catheter placement is not possible.
- When provocative testing indicates intolerance to the occlusion procedure.
- When vasospasm stops blood flow.

Precautions: 1) The safety and effectiveness has not been studied in the following patient populations: pregnant and nursing women, individuals less than 18 years old, individuals with aneurysms not associated with a bAVM nidus, or distal feeders to a bAVM nidus or dural AV fistulas. 2) Some data indicate that dimethyl sulfoxide potentiates other concomitantly administered medications. 3) A garlic-like taste may be noted by the patient with use of the Onyx™ LES due to the DMSO component. This taste may last several hours. An odor on the breath and skin may be present. 4) Inspect product packaging prior to use. Do not use if sterile barrier is open or damaged. 5) Use prior to expiration date. 6) Verify that the catheters and accessories (see directions for use) used in direct contact with the Onyx™ LES polymer are clean and compatible with the material and do not trigger polymerization or degrade with contact. Use only ev3 approved, Onyx™ LES/DMSO compatible micro catheters indicated for use in the neurovasculature and ev3 syringes. Other micro catheters or syringes may not be compatible with DMSO and their use can result in thromboembolic events due to catheter degradation. Refer to the Warnings and Directions for Use sections. 7) Wait a few seconds following completion of the Onyx™ LES injection before attempting catheter retrieval. Failure to wait a few seconds to retrieve the micro catheter after the Onyx™ LES injection may result in fragmentation of the Onyx™ LES into non-target vessels.

Difficult catheter removal or catheter entrapment may be caused by any of the following: Angioarchitecture: very distal bAVM fed by afferent, lengthened, small, or tortuous pedicles, Vasospasm, Reflux, Injection time. To reduce the risk of catheter entrapment, carefully select catheter placement and manage reflux to minimize the factors listed above.

Should catheter removal become difficult, the following will assist in catheter retrieval: Carefully pull the catheter to assess any resistance to removal. If resistance is felt, remove any "slack" in the catheter. Gently apply traction to the catheter (approximately 3-4 cm of stretch to the catheter). Hold this traction for a few seconds and release.

Assess traction on vasculature to minimize risk of hemorrhage. This process can be repeated intermittently until catheter is retrieved.

Alternate Technique for Difficult to Remove Catheters: Remove all slack from the catheter by putting a few centimeters of traction on the catheter to create a slight tension in the catheter system. Firmly hold the catheter and then pull it using a quick wrist snap motion (from left to right) 10 – 15 centimeters to remove the catheter from the Onyx™ LES cast (Note: Do not apply more than 20 cm of traction to catheter, to minimize risk of catheter separation).

For entrapped catheters: Under some difficult clinical situations, rather than risk rupturing the malformation and consequent hemorrhagic complications by applying too much traction on an entrapped catheter, it may be safer to leave the micro catheter in the vascular system. This is accomplished by stretching the catheter and cutting the shaft near the entry point of vascular access allowing the catheter to remain in the artery. If the catheter breaks during removal, distal migration or coiling of the catheter may occur. Same day surgical resection should be considered to minimize the risk of thrombosis.

Potential Complications: The following adverse events occurred using Onyx™ during a prospective, randomized, multi-center clinical trial for the presurgical treatment of bAVMs: Death, Headache +/- nausea and vomiting, Patient discomfort, Laboratory/Imaging abnormalities (Endocrine/Metabolic, Hematologic, Asymptomatic MRI/CT Findings, Respiratory/Pulmonary, General, Gastrointestinal (GI)), Worsening Neurologic Status (Persistent, Resolved), Hyperglycemia, Infection, Bleeding and/or Low Hct requiring transfusion (Surgical Bleeding, Decreased Hct Requiring Transfusion), Intracranial Hemorrhage, Medication reaction, Failed access, Access site bleeding, Fever, Delivery Catheter removal difficulty, Poor penetration/visualization, Hypotension, Stroke, Cardiac arrhythmia, Hydrocephalus, SIADH (Syndrome of inappropriate antidiuretic hormone secretion, dilutional hyponatremia), Vessel Dissection, Hypertension, Limb ischemia, Respiratory failure, Seizures, UTI (Urinary tract infection), Vasospasm, Vaso-vagal episode, catheter shaft rupture, delivery catheter rupture, fragmentation of the Onyx™ LES, hypoxia, laryngospasm, peptic ulcer disease, psychotic episode, pulmonary edema, skin abrasion, subintimal injection, tachypnea, and tongue swelling.

Additional adverse events, which may be associated with embolization procedures include: Allergic reaction, Thrombocytopenia, Pulmonary embolism, Catheter entrapment, Catheter rupture, Device migration and cast movement, Hemorrhagic complications related to attempts to remove entrapped catheter.

WARNINGS: Serious, including fatal, consequences could result with the use of the Onyx™ LES without adequate training. Contact your Medtronic sales representative for information on training courses.

Complete indications, Contraindications, warnings and instructions for use can be found in the product labeling supplied with each device.

INDICATIONS FOR USE

Axium™ and Axium™ Prime detachable coils

The Axium™ and Axium™ Prime detachable coils are not intended for all patients and may not be the appropriate treatment for all clinical scenarios. Axium™ and Axium™ Prime detachable coils are intended for the endovascular embolization of intracranial aneurysms. Axium™ and Axium™ Prime detachable coils are also intended for the embolization of other neurovascular abnormalities such as arteriovenous malformations and arteriovenous fistulae. Axium™ Prime Detachable Coil (Frame): The Axium™ Prime detachable coil system is indicated for the endovascular embolization of intracranial aneurysms and other neurovascular abnormalities, such as arteriovenous malformations and arteriovenous fistulae. The Axium™ Prime detachable coils are also indicated for arterial and venous embolizations in the peripheral vasculature.

Pipeline™ Flex embolization device

The Pipeline™ Flex embolization device should be used only by physicians trained in percutaneous, intravascular techniques and procedures at medical facilities with the appropriate fluoroscopy equipment.

Indications for Use: The Pipeline™ Flex embolization device is indicated for the endovascular treatment of adults (22 years of age or older) with large or giant wide-necked intracranial aneurysms (IAs) in the internal carotid artery from the petrous to the superior hypophyseal segments.

CAUTION: Federal (USA) law restricts this device to sale, distribution and use by or on the order of a physician. Indications, contraindications, warnings and instructions for use can be found in the product labeling supplied with each device.

Warnings: 1) Resheathing of the Pipeline™ Flex embolization device more than 2 full cycles may cause damage to the distal or proximal ends of the braid. 2) Persons with known allergy to platinum or cobalt/chromium alloy (including the major elements platinum, cobalt, chromium, nickel, molybdenum or tungsten) may suffer an allergic reaction to the Pipeline™ Flex embolization device implant. 3) Person with known allergy to tin, silver, stainless steel or silicone elastomer may suffer an allergic reaction to the Pipeline™ Flex embolization device delivery system. 4) Do not reprocess or resterilize. Reprocessing and resterilization increase the risk of patient infection and compromised device performance. 5) Delayed rupture may occur with large and giant aneurysms. 6) Placement of multiple Pipeline™ Flex embolization devices may increase the risk of ischemic complications.

Precautions: 1) Do not use product if the sterile package is damaged. 2) Do not use the Pipeline™ Flex embolization device in patients in whom angiography demonstrates inappropriate anatomy, such as severe pre- or post-aneurysmal narrowing. 3) The Pipeline™ Flex embolization device should be used only by physicians trained in percutaneous, intravascular techniques and procedures at medical facilities with the appropriate fluoroscopic equipment. 4) Physicians should undergo appropriate training prior to using the Pipeline™ Flex embolization device in patients. 5) The Pipeline™ Flex embolization device is provided sterile for single use only. Store in a cool, dry place. 6) Carefully inspect the sterile package and device components prior to use to verify that they have not been damaged during shipping. Do not use kinked or damaged components. 7) Use the Pipeline™ Flex embolization device system prior to the "Use By" date printed on the package. 8) The appropriate anti-platelet and anticoagulation therapy should be administered in accordance with standard medical practice. 9) A thrombosing aneurysm may aggravate pre-existing, or cause new, symptoms of mass effect and may require medical therapy. 10) Do not attempt to reposition after deployment. 11) Do not use in patients in whom the angiography demonstrates the anatomy is not appropriate for endovascular treatment, due to conditions such as severe intracranial vessel tortuosity or stenosis. 12) Use of implants with labeled diameter larger than the parent vessel diameter may result in decreased effectiveness and additional safety risk due to incomplete foreshortening resulting in an implant longer than anticipated.

Potential Complications: Potential complications, some of which could be fatal, include, but are not limited to the following: Adverse reaction to antiplatelet/ anticoagulation agents or contrast media; Blindness; Coma; Device fracture; Device migration or misplacement; Dissection of the parent artery; Embolism; Groin injury; Headache; Hemorrhage; Hydrocephalus; Infection; Intracerebral bleeding; Ischemia; Mass effect; Neurological deficits; Parent Artery Stenosis; Perforation; Perforator occlusion; Post- procedure bleeding; Ruptured or perforated aneurysm; Seizure; Stroke; Thromboembolism; Transient Ischemic Attack (TIA); Vasospasm; Vessel occlusion; Vessel perforation; Vision impairment.

Contraindications: The use of the Pipeline™ Flex embolization device is contraindicated for patients with any of the following conditions:

- 1) Patients with active bacterial infection.
- 2) Patients in whom dual antiplatelet therapy (aspirin and clopidogrel) is contraindicated.
- 3) Patients who have not received dual antiplatelet agents prior to the procedure.
- 4) Patients in whom a pre-existing stent is in place in the parent artery at the target aneurysm location.

Solitaire™ Platinum revascularization device

The Solitaire™ Platinum revascularization device is intended to restore blood flow by removing thrombus from a large intracranial vessel in patients experiencing ischemic stroke within 8 hours of symptom onset. Patients who are ineligible for intravenous tissue plasminogen activator (IV t-PA) or who fail IV t-PA therapy are candidates for treatment.

The Solitaire™ revascularization device is indicated for use to restore blood flow in the neurovasculature by removing thrombus for the treatment of acute ischemic stroke to reduce disability in patients with a persistent, proximal anterior circulation, large vessel occlusion, and smaller core infarcts who have first received intravenous tissue plasminogen activator (IV t-PA). Endovascular therapy with the device should be started within 6 hours of symptom onset.

CAUTION: Federal (USA) law restricts this device to sale distribution and use by or on order of a physician. Indications, contraindications, warnings and instructions for use can be found in the product labeling supplied with each device. The Solitaire™ Platinum revascularization device should only be used by physicians trained in interventional neuroradiology and treatment of ischemic stroke. Carefully inspect the sterile package and the Solitaire™ Platinum revascularization device prior to use to verify that neither has been damaged during shipment. Do not use kinked or damaged components. The Solitaire™ Platinum revascularization device is not to be used after the expiration date imprinted on the product label. Refer to the appropriate intravenous tissue plasminogen activator (IV t-PA) manufacturer labeling for indications, contraindications, warnings, precautions, and instructions for use.

CONTRAINDICATIONS

Use of the Solitaire™ Platinum revascularization device is contraindicated under these circumstances. • Patients with known hypersensitivity to nickel-titanium. Patients with stenosis and/or pre existing stent proximal to the thrombus site that may preclude safe recovery of the Solitaire™ Platinum revascularization device. Patients with angiographic evidence of carotid dissection. • **POTENTIAL COMPLICATIONS** • Possible complications include, but are not limited to, the following: Hematoma and hemorrhage at puncture site; Perforation or dissection of the vessel; Vasoconstriction (Vasospasm); Change in mental status; Persistent neurological deficits; Neurologic deterioration including stroke progression, stroke in new vascular territory, and death; Brain Edema; Ischemia; Inflammation; Infection; Allergic reactions; Air Embolism Intracranial Hemorrhage; Vascular occlusion; Pseudo aneurysm formation; Post procedure bleeding; Distal embolization including to a previously uninvolved territory; Adverse reaction to antiplatelet/anticoagulation agents or contrast media; Device(s) deformation, collapse, fracture or malfunction Thrombosis (acute and subacute); Arteriovenous Fistula.

INDICATIONS FOR USE

WARNINGS — BOTH INDICATIONS

The appropriate anti-platelet and anti-coagulation therapy should be administered in accordance with standard medical practice. • Per IV t-PA manufacturer labeling, IV t-PA should be administered within 3 hours of stroke symptom onset (IV t-PA use beyond 3 hours is not approved in the United States). • Do not torque the Solitaire™ Platinum revascularization device. • For vessel safety, do not perform more than three recovery attempts in the same vessel using Solitaire™ Platinum revascularization devices. • For device safety, do not use each Solitaire™ Platinum revascularization device for more than two flow restoration recoveries. • For each new Solitaire™ Platinum revascularization device, use a new microcatheter. • Solitaire™ Platinum revascularization device does not allow for electrolytic detachment. • To prevent device separation: Do not oversize device; Do not recover (i.e. pull back) the device when encountering excessive resistance. Instead, resheath the device with the microcatheter and then, remove the entire system under aspiration. If resistance is encountered during resheathing, discontinue and remove the entire system under aspiration; Do not treat patients with known stenosis proximal to the thrombus site. • This device is supplied STERILE for single use only. Do not reprocess or re-sterilize. Reprocessing and re-sterilization increase the risks of patient infection and compromised device performance. • If excessive resistance is encountered during the delivery of the Solitaire™ Platinum revascularization device, discontinue the delivery and identify the cause of the resistance. Advancement of the Solitaire™ Platinum revascularization device against resistance may result in device damage and/or patient injury. • If excessive resistance is encountered during recovery of the Solitaire™ Platinum revascularization device, discontinue the recovery and identify the cause of the resistance. • For vessel safety, do not perform more than three recovery attempts in the same vessel using the Solitaire™ Platinum revascularization device. • For device safety, do not use each Solitaire™ Platinum revascularization device for more than two flow restoration recoveries. • Advancing the microcatheter while the device is engaged in clot may lead to embolization of debris. • Do not advance the microcatheter against any resistance. • Do not reposition more than two times.

WARNINGS — INDICATION 1 ONLY

The safety and effectiveness has not been established for the Solitaire™ Platinum device to reduce disability in patients with the following: Posterior circulation occlusions; More distal occlusions in the anterior circulation; Large core infarct (ASPECTS ≤ 7).

Riptide™ aspiration system

The Riptide™ Aspiration System is intended for use in the revascularization of patients with acute ischemic stroke secondary to intracranial large vessel occlusive disease (within the internal carotid, middle cerebral M1 and M2 segments, basilar, and vertebral arteries) within 8 hours of symptom onset. Patients who are ineligible for intravenous tissue plasminogen activator (IV t-PA) or who fail IV t-PA therapy are candidates for treatment.

Cadence™ precision injector

The Cadence precision injector is intended for the controlled delivery of fluids in the inflation and deflation of temporary occlusion balloons.

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